

Youran Wang

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Education

University of Wisconsin–Madison

B.S. in Computer Science (Honors) and B.S. in Mathematics

Madison, Wisconsin

Sep 2024 – Dec 2026 (Expected)

- GPA: 3.978/4.000, Dean's List (All Semesters)
- Advisor: Prof. Frederic Sala
- Selected courses: Foundation Models, Integer Optimization, Big Data Systems, Database Management Systems, Algorithms, Linear Optimization, Topology, Real Analysis, Probability Theory, Honors Thesis I & II

Research Interests

My research interests lie in robot learning and machine learning for physical systems, especially how robots can learn representations and models of the world that let them act reliably under uncertainty. My recent work studies belief-space planning for interactive driving, safe manipulation, and representation learning with foundation models.

Publications and Manuscripts

PREPRINTS / UNDER REVIEW

[1] **Youran Wang**[†], Devodita Chakravarty[†], and Yiwei Lyu*, “When to Probe and When to Act: Clarity-Aware Risk-Adaptive Control for Interactive Driving,” *Under Review at ICRA 2027*. ([†]Equal contribution)

[2] Sonia Crompt, Satya Sai Srinath Namburi GNVV, **Youran Wang**, Grace Kisslinger, Frederic Sala, James Booth, and Allegra LeGrande, “Geospatial Diffusion-based Evolution Synthesis (GeoDES) for Storm-Centered Weather Augmentation,” *Under Review at ICLR 2027*. [[Paper](#)]

Research Experience

Robotics Lab, Texas A&M University

Research Assistant, Advised by Prof. Yiwei Lyu

Mar 2026 – Sep 2026

College Station, Texas

- **Interactive driving:** Developed a belief-space MPC framework under unknown human driving styles. Introduced a scalar *clarity* state, derived from the predicted belief covariance, that jointly drives active probing of the latent driver characteristic and adapts the conservatism of a CVaR barrier, so the planner probes under high ambiguity and shifts to task execution as confidence grows, without explicit mode switching.
- **Safe manipulation:** Developed a framework that learns object properties from purposeful nudges and incorporates the resulting belief into physics-informed control barrier functions to adapt pushing trajectories. Evaluated safety and goal-reaching performance during deployment in Isaac Lab.

University of Wisconsin–Madison Sprocket Lab

Research Assistant, Advised by Prof. Frederic Sala

Nov 2025 – Present

Madison, Wisconsin

- Trained GeoDES, a phenomenon-centered spatiotemporal diffusion model for synthesizing rare geospatial events, via 2D spatial pretraining, temporal inflation, and 3D fine-tuning on multivariable ERA5 storm data.
- GeoDES reduces peak vorticity error by 21% and improves synoptic ACC by 26% over the strongest baseline on North Atlantic cyclones, and generalizes to post-disaster satellite imagery.

Stony Brook University

Research Assistant, Advised by Prof. Chenyu You

May 2026 – Present

Remote

- Developing structured tokenization for EEG foundation models to separate physiological and temporal features.
- Studying cross-task generalization and subject-specific bias in EEG foundation models.

University of Wisconsin–Madison Mathematics Research Program

Madison Experimental Mathematics Lab, Advised by Prof. Chenxi Wu

Spring 2026

Madison, Wisconsin

- Implemented kneading-theory algorithms for topological entropy of post-critically finite polynomial maps.
- Analyzed entropy spectra and Galois conjugates across polynomial families.

Teaching and Service

Math Grader

Fall 2026

Department of Mathematics, University of Wisconsin–Madison

Madison, Wisconsin

- Grade exams and quizzes for proof-based Linear Algebra (Math 341).

Math Learning Center Tutor

Sep 2025 – May 2026

Course Assistant & Peer Mentor, University of Wisconsin–Madison

Madison, Wisconsin

- Served as Course Assistant for proof-based mathematics courses, including Linear Algebra (Math 341) and Number Theory (Math 467), leading office hours and mentoring students on proof techniques and rigorous problem-solving.
- Provided in-class academic support as a Peer Mentor for precalculus-level courses.

Work Experience and Projects

Application Development Intern

Summer 2025

Roundtable Strategy

Boston, Massachusetts

- Collaborated with senior engineers to design, build, and deploy scalable distributed applications (desktop & mobile) using Python and JavaScript. Integrated AI libraries into high-performance data pipelines and automated workflows.
- Optimized deployment on cloud-based infrastructure, reducing system latency by 30% and improving throughput and reliability for client-facing solutions.

Summer of AI Laboratory

Summer 2025

SAIL Program, University of Wisconsin–Madison

Madison, Wisconsin

- Designed and deployed an AI-enabled distributed simulation platform for a trading card game, supporting concurrent multi-user interactions and scalable backend services.
- Built low-latency inference pipelines to power real-time strategy generation, integrating LLM agents with dynamic game state processing.

Technical Skills

Programming Languages: Python, C/C++, Java, MATLAB, R, JavaScript.

Machine Learning: PyTorch, scikit-learn, foundation models, representation learning, reinforcement learning.

Robotics and Control: Isaac Sim, Isaac Lab, MuJoCo, ROS, control barrier functions.

Development Tools: Git, Linux, Jupyter, LaTeX

Languages: English, Chinese